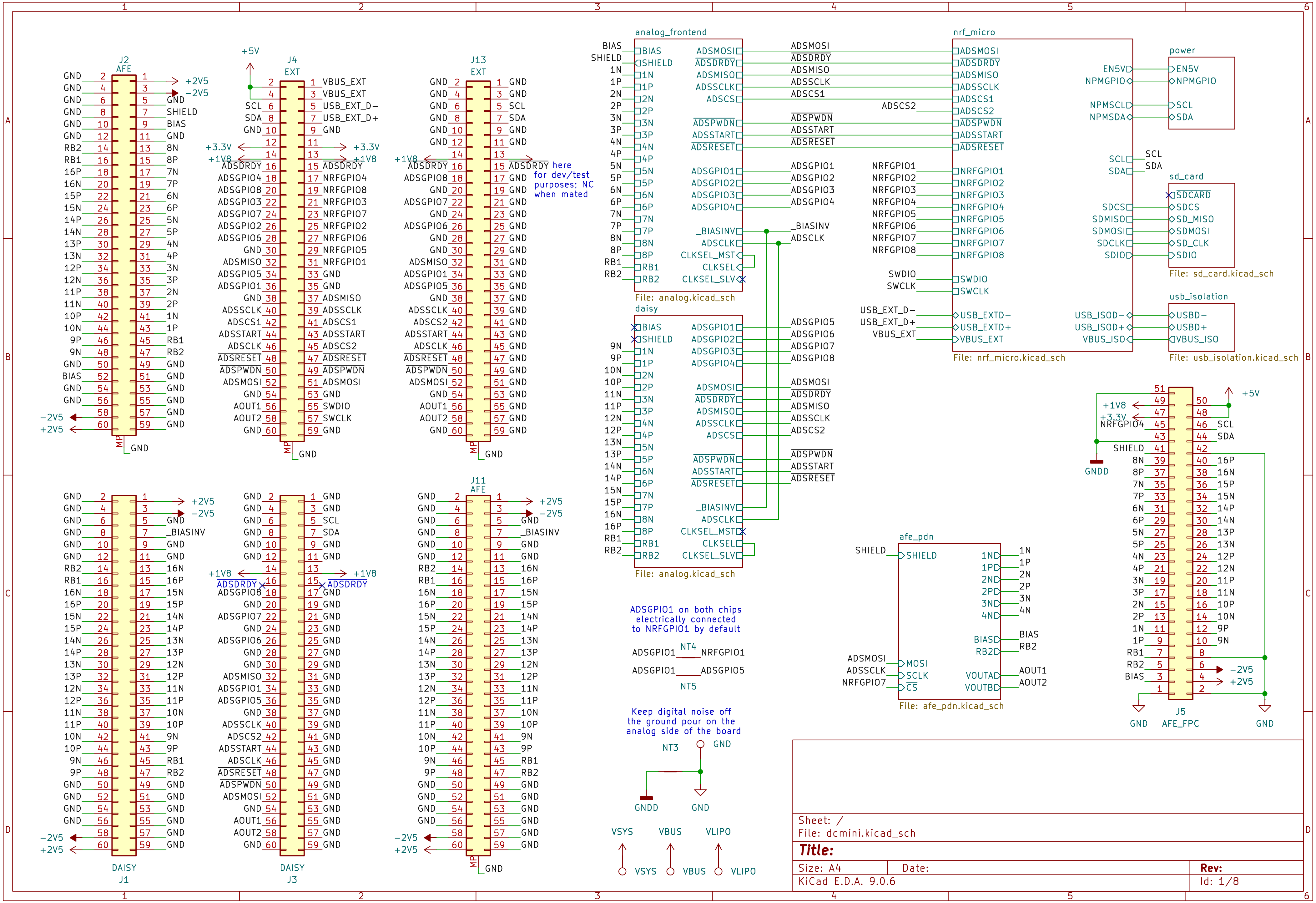
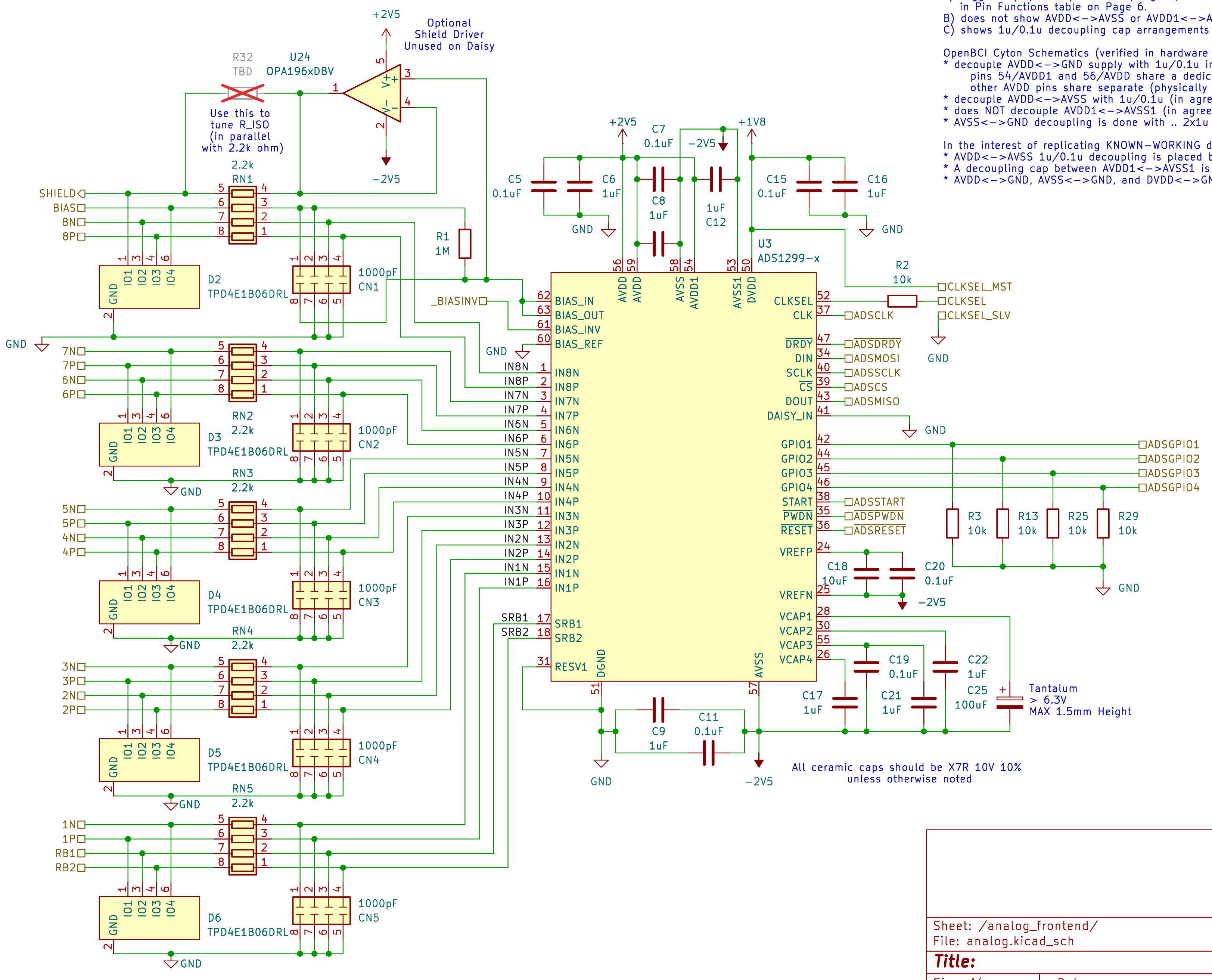


J2
AFE
J4
EXT
J13
EXT
J1
DAISY
J3
DAISY
J11
AFE
J5
AFE_FPC
analog_frontend
File: analog.kicad_sch
daisy
File: analog.kicad_sch
nrf_micro
File: nrf_micro.kicad_sch
power
sd_card
File: sd_card.kicad_sch
usb_isolation
File: usb_isolation.kicad_sch
afe_pdn
File: afe_pdn.kicad_sch
ADSDRDY here for dev/test purposes; NC when mated
ADSGPIO1 on both chips electrically connected to NRFGPIO1 by default
ADSGPIO1 NT4 NRFGPIO1
ADSGPIO1 NT5 ADSGPIO5
Keep digital noise off the ground pour on the analog side of the board
NT3
GND
GNDD
VSYS
VBUS
VLIPO
+5V
+3.3V
+1V8
+2V5
−2V5
Sheet: /
File: dcmini.kicad_sch
Title:
Size: A4
Date:
KiCad E.D.A. 9.0.6
Rev:
Id: 1/8

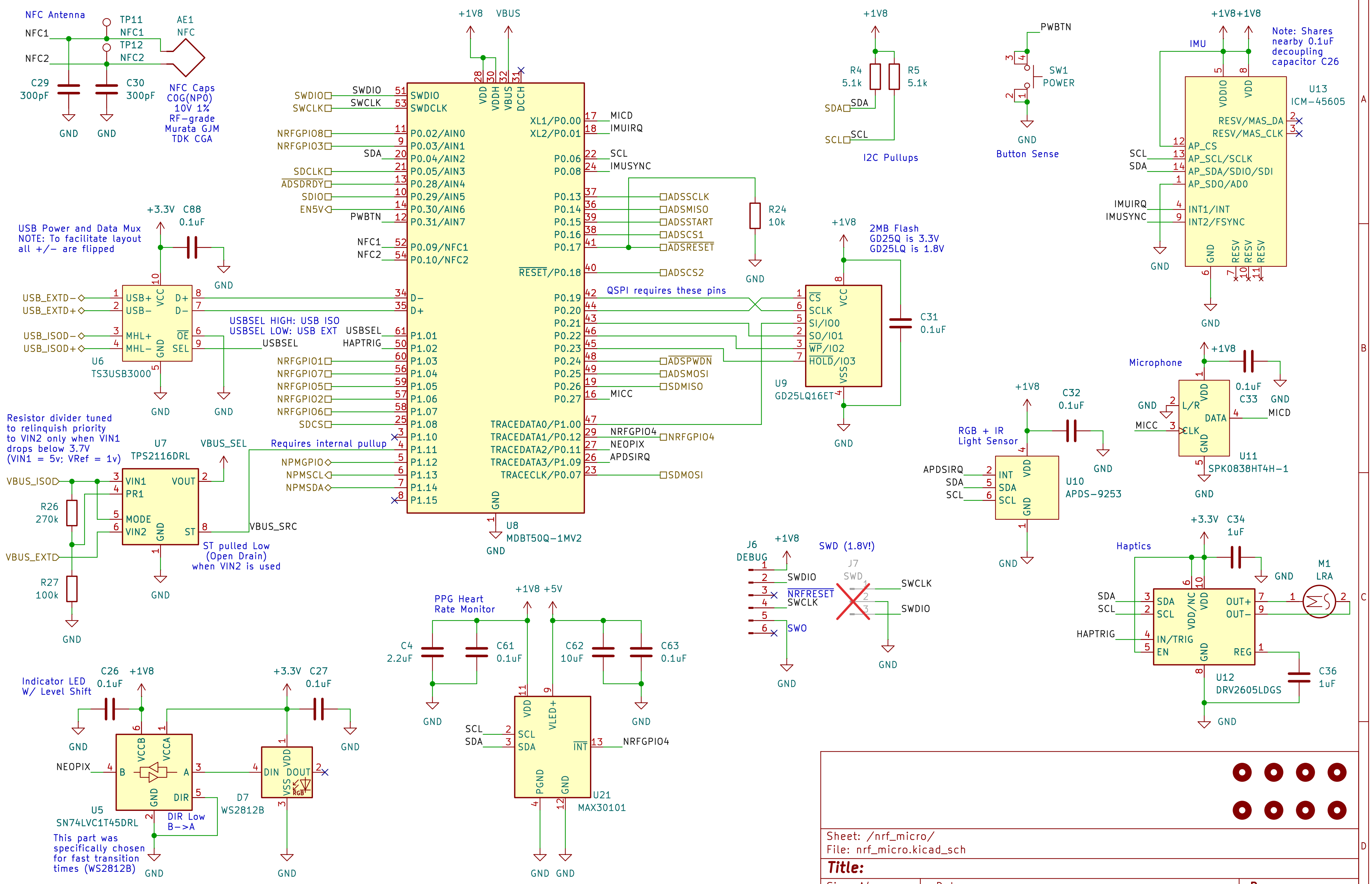
ADS1299 Datasheet gives conflicting decoupling advice for a bipolar supply config (+/-2.5V)
A) suggests (separate?) 1u decoupling caps between AVDD1<->AVSS1 AND AVDD<->AVSS in Pin Functions table on Page 6.
B) does not show AVDD<->AVSS or AVDD1<->AVSS1 decoupling at all for bipolar supply in Figure 78.
C) shows 1u/0.1u decoupling cap arrangements for supply decoupling in Figure 78

OpenBCI Cyton Schematics (verified in hardware with bipolar supply)
* decouple AVDD<->GND supply with 1u/0.1u in TWO separate (nearby) locations (?)
  pins 54/AVDD1 and 56/AVDD share a dedicated 1u/0.1u decoupling to GND
  other AVDD pins share separate (physically nearby) 1u/0.1u decoupling to GND
* decouple AVDD<->AVSS with 1u/0.1u (in agreement with advice A and C, but not B)
* does NOT decouple AVDD1<->AVSS1 (in agreement with advice B but not advice A)
* AVSS<->GND decoupling is done with .. 2x1u (2u) caps (in error??)

In the interest of replicating KNOWN-WORKING design (OpenBCI Cyton) while following advice
* AVDD<->AVSS 1u/0.1u decoupling is placed between pins 58/59
* A decoupling cap between AVDD1<->AVSS1 is present, may improve noise
* AVDD<->GND, AVSS<->GND, and DVDD<->GND are all decoupled exactly once with a 1u and 0.1u cap



Sheet: /analog_frontend/
File: analog.kicad_sch
Title:
Size: A4 | Date: | Rev:
KiCad E.D.A. 9.0.6 | Id: 2/8

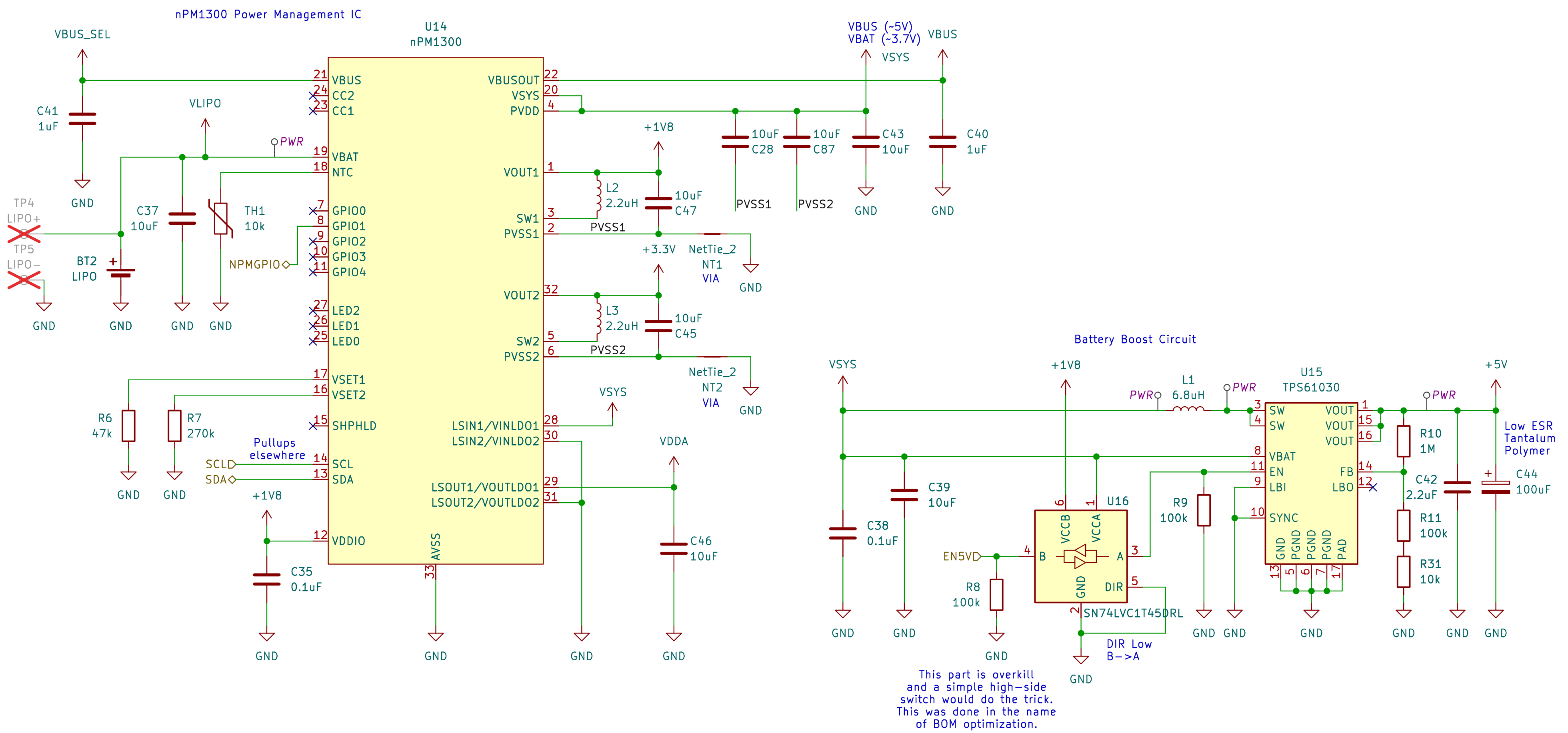


Sheet: /nrf_micro/
File: nrf_micro.kicad_sch

Title:

Size: A4 | Date: | Rev:

KiCad E.D.A. 9.0.6 | Id: 3/8

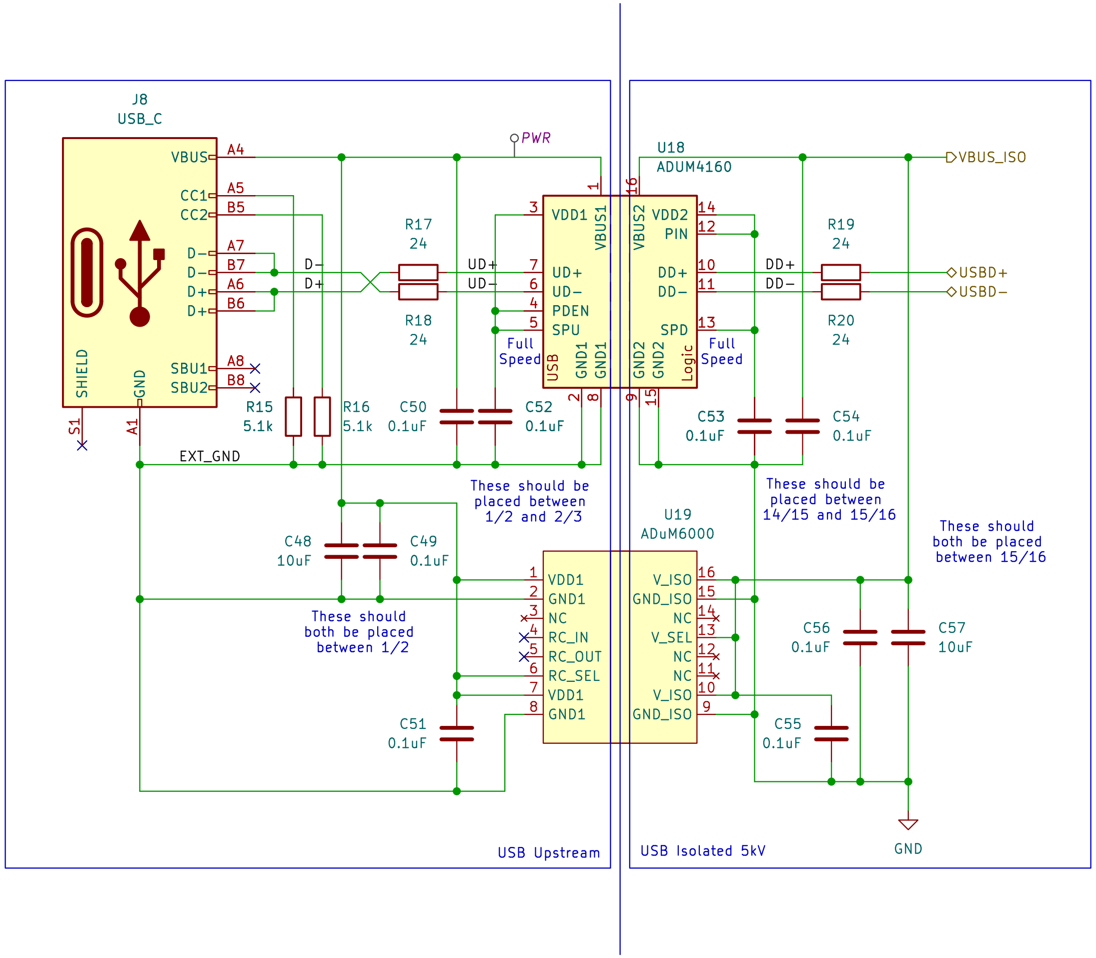


Sheet: /power/
File: power.kicad_sch

Title:

Size: A4 | Date: | Rev:

KiCad E.D.A. 9.0.6 | Id: 4/8



ADuM6000 provides MAXIMUM 80 mA of current.
It gets VERY hot when doing so.

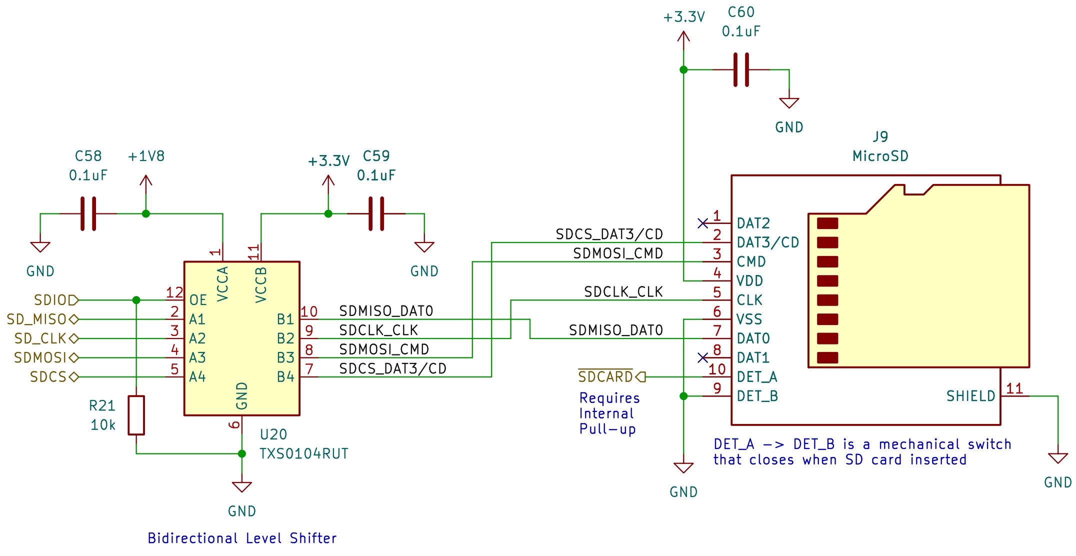
Recommend keeping I_load from VBUS at maximum 50mA.
External hardware connected to supplied power rails should
be tuned to keep current draw limited, or else ADuM6000
could get hot enough to damage battery if in proximity

Schematic based on Adafruit USB Isolator
https://github.com/adafruit/Adafruit-USB-Isolator-PCB
Substituted ADUM6000 in place of ADUM5000
ADUM6000 provides 80mA/5kV isolation instead of 100mA/2.5kV
Limited to USB Full-Speed (12 Mbps)
Limited to ~80ma supply current on VBUS

Sheet: /usb_isolation/
File: usb_isolation.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 5/8



Sheet: /sd_card/
File: sd_card.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 9.0.6 | Id: 6/8

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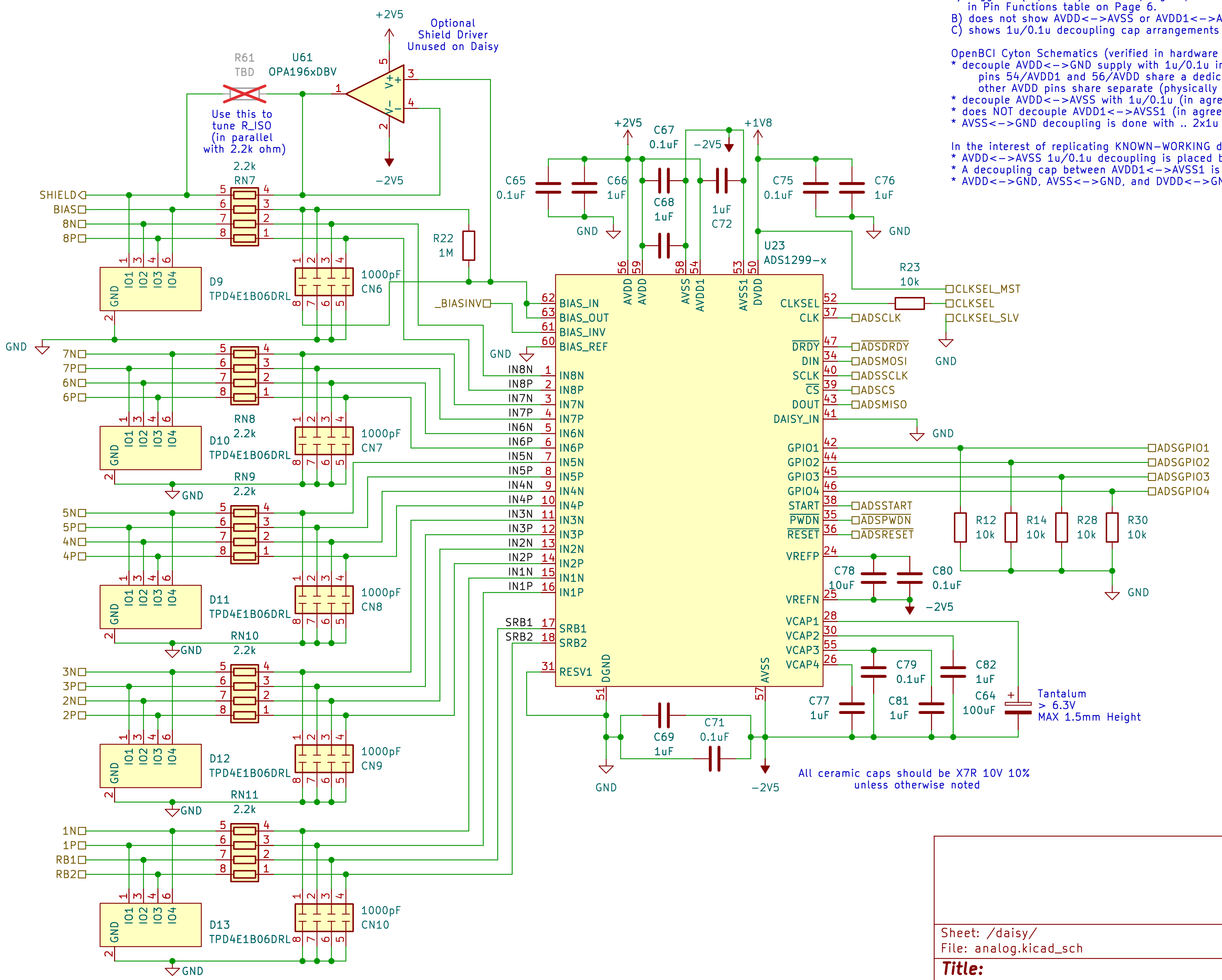
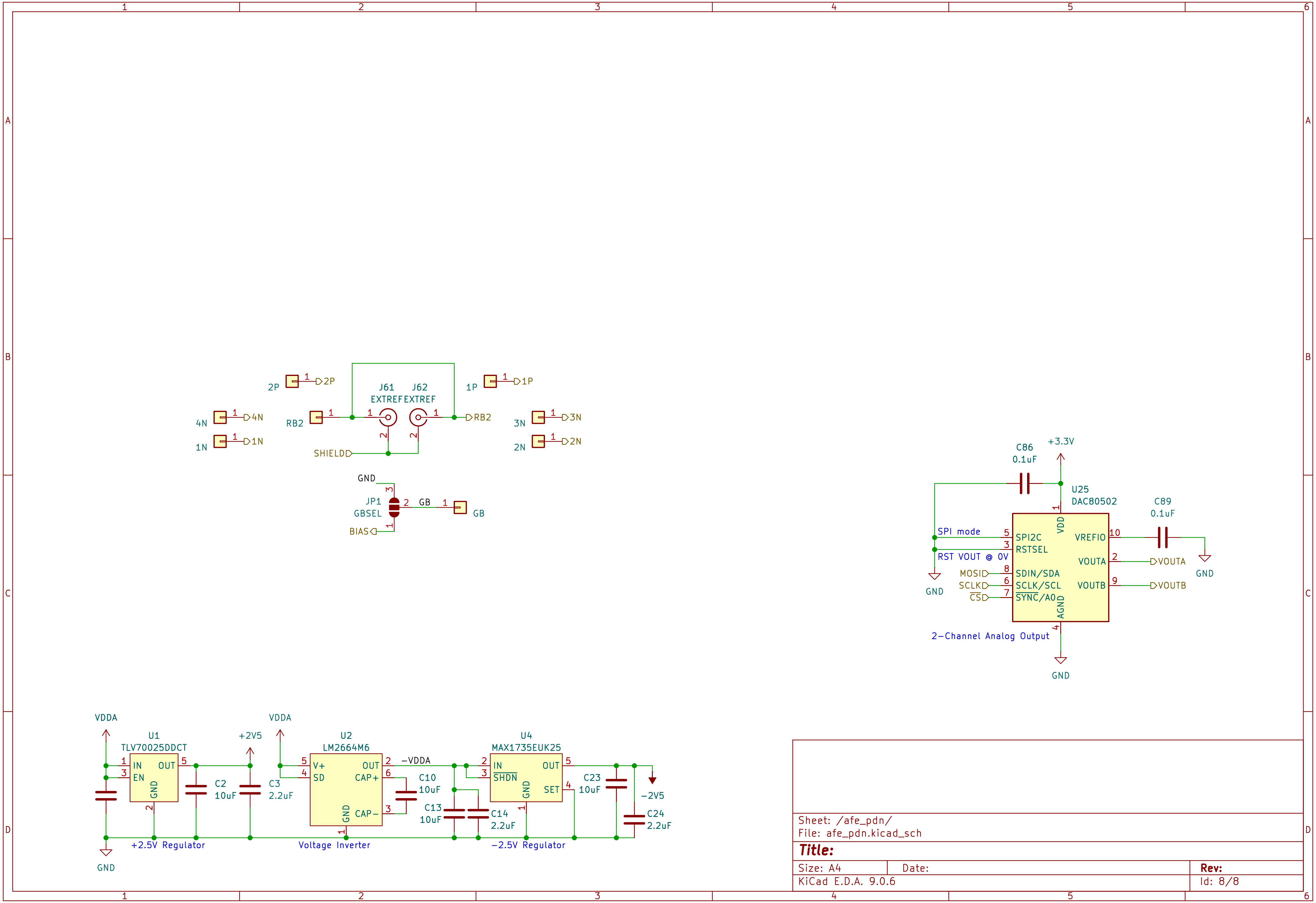


Sheet: /daisy/
File: analog.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 9.0.6 | Id: 7/8


2P
1P
J61
J62
EXTREF
EXTREF
4N
1N
RB2
SHIELD
3N
2N
GND
JP1
GBSEL
GB
BIAS
C86
0.1uF
+3.3V
U25
DAC80502
C89
0.1uF
SPI mode
SPI2C
RSTSEL
RST VOUT @ 0V
VDD
VREFIO
VOUTA
VOUTB
MOSI
SCLK
CS
SDIN/SDA
SCLK/SCL
SYNC/A0
AGND
2-Channel Analog Output
VDDA
U1
TLV70025DDCT
IN
EN
OUT
C2
10uF
C3
2.2uF
+2V5
+2.5V Regulator
U2
LM2664M6
V+
SD
OUT
CAP+
CAP-
-VDDA
C10
10uF
C13
10uF
C14
2.2uF
Voltage Inverter
U4
MAX1735EUK25
SHDN
SET
C23
10uF
C24
2.2uF
-2V5
-2.5V Regulator
Sheet: /afe_pdn/
File: afe_pdn.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.6
Id: 8/8